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MULTI-PRODUCT FIRMS AND PRODUCT SWITCHING

by

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Abstract

This paper examines the frequency, pervasiveness and determinants of product switching by U.S. manufacturing firms. We find that one-half of firms alter their mix of five-digit SIC products every five years, that product switching is correlated with both firm- and firm-product attributes, and that product adding and dropping induce large changes in firm scope. The behavior we observe is consistent with a natural generalization of existing theories of industry dynamics that incorporates endogenous product selection within firms. Our findings suggest that product switching contributes to a reallocation of resources within firms towards their most efficient use.

Keywords: Heterogeneous firms, Product differentiation, Product market entry and exit,

JEL classification: D21, E23, L11, L60

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1. Introduction

The extent to which resources are allocated to their best use is a core issue of economics. Until now, research into industry dynamics has addressed this issue by focusing almost exclusively on the contribution of firm entry and exit to resource reallocation, that is, whether newly created firms or plants are more productive than the dying firms and plants they replace.[1] This paper examines a new, "extensive" margin of firm adjustment, the reassignment of resources that takes place within surviving firms as they add and drop (i.e., "switch") products.

Our analysis of product switching makes use of a unique longitudinal dataset that tracks U.S. firms' manufacturing output across products between quinquennial U.S. Manufacturing Censuses from 1987 to 1997. In this dataset, a "product" is one of approximately fifteen hundred five-digit Standard Industrial Classification (SIC) categories, e.g., "Passenger Cars".[2] We observe the set of products each manufacturing firm produces in each census year and analyze how incumbent firms' mix of products evolves from one census year to the next. To our knowledge, these are the most comprehensive data on multi-product production yet assembled. Standard manufacturing censuses, for example, typically record just the primary industry of each establishment.[3]

We find product switching to be frequent, widespread and influential in determining both aggregate and firm outcomes. On average, recently added and about-to-be dropped products each account for around one sixth of a product's output, amounts that rival the shares represented by recently created and about-to-exit firms. At the firm level, we find that more than one half of U.S. manufacturing firms alter their mix of products between censuses, and that one half of those firms change their mix of products by *both* adding and dropping at least one product

[1] There is a large empirical literature in macroeconomics on firm creation and destruction and its implications for industry dynamics and the firm-size distribution. See, for example, Albuquerque and Hopenhayn (2002), Baily et al. (1992), Cooley and Quadini (2001), Dunne et al. (1989a,b), Foster et al. (2001, 2008), and Rossi-Hansberg and Wright (2007) among others.

[2] Throughout the paper, we refer to two-, four- and five-digit SIC categories as "sectors", "industries" and "products", respectively.

[3] Existing empirical work on multiple-product firms typically examines product diversification at a point in time. See, for example, Gollop and Monahan (1991) and Baldwin and Gu (2005). Dunne et al. (1988, 1989b) examine product diversification as a mode of market entry distinct from plant birth, while Dunne et al. (2005) investigate the empirical relationship between the mode of market entry and plant death.

every five years. Product adding and dropping also exert considerable influence on the scope of firms, with an average of 40 percent of firms adding products outside their existing set of four-digit SIC industries between census years. Given the unobserved changes firms presumably make to their product mix at lower levels of aggregation, our estimates of product switching likely underestimate the true importance of firms' adjustments to their extensive margins.[4]

Our empirical analysis is guided by a model of endogenous product selection that builds on existing theories of industry dynamics by Jovanovic (1982), Hopenhayn (1992), Ericson and Pakes (1995), Melitz (2003) and Bernard, Redding and Schott (2007).[5] In these existing models, firms that are heterogeneous in productivity are assumed to produce a single product, with the result that firm and product-market entry and exit are equivalent. Here, we develop a natural extension of such models in which firms choose to produce an endogenous range of products in response to evolving firm and firm-product characteristics.[6] In our model, firms differ in innate productivity while firms' products vary in their attractiveness to consumers *vis a vis* other producers of the same product. The overall profitability of a firm depends on the interaction of these attributes. Higher values of consumer tastes for a firm's product raise the firm's profitability in that product, while higher values of firm productivity increase a firm's profitability in *all* products. In equilibrium, the most productive firms manufacture the largest ranges of products because they earn greater revenue per product for given values of consumer tastes and can therefore cover the fixed costs of a wider set of products.

Our framework provides a basis for understanding many of the empirical regularities discernible in the census data. In the data, we find that multi-product firms exhibit higher productivity than single-product firms. In the model, this is due to high-productivity firms'

[4] One of the attractions of our data is that information is available for the entire manufacturing sector. Scanner data, such as those used by Chevalier et al. (2003) and Broda and Weinstein (2007), offer the potential to measure products at finer levels of disaggregation, although they are typically only available for specific categories of goods.

[5] These models receive empirical support from studies of firm creation and destruction by Baily et al. (1992), Dunne et al. (1989a,b) and Foster et al. (2001, 2006).

[6] Existing theoretical research on multiple-product firms focuses on issues associated with managing a given range of products at a particular point in time (e.g. Chandler 1920, Baumol 1977, Panzar and Willig 1977, Brander and Eaton 1984, Shaked and Sutton 1990 and Eaton and Schmitt 1994). More recently, Klette and Kortum (2004), Luttmer (2007) and Chatterjee and Rossi-Hansberg (2008) have explored the role of innovation in determining firm scope, as discussed further below.

ability to cover the fixed costs of a greater number of products. In the data, we find product switching and firm creation and destruction to be both frequent and widespread across sectors. In the model, an interaction of stochastic shocks to firm productivity and stochastic shocks to consumer tastes fosters steady-state product adding and dropping as well as steady-state firm creation and destruction. In the data, we observe a positive correlation between products' add and drop rates. In the model, this correlation arises because firms receiving positive demand shocks add the product at the same time that some incumbent producers receiving negative demand shocks drop it. Likewise, in the data we find that the probability that a firm drops a product declines with firm-product shipments and firm-product tenure. In the model, this scale and age dependence arises as a result of serial correlation in idiosyncratic shocks to firm-product profitability. Firms' lower-volume and recently-added products are more likely to be dropped as a result of a negative shock.

Our model also offers insight into the importance of firms for product entry and exit. In the data, we find that a firm's decision to subsequently add a new product is positively correlated with its *ex ante* productivity in its existing set of products. Similarly, we find that a firm's decision to drop a product is influenced by firm as well as firm-product characteristics. In the model, such an interdependence of firm and firm-product attributes results from the fact that higher (firm) productivity raises the profitability of all products the firm might produce. We also find that product switching is correlated with contemporaneous changes in firm outcomes. In the data, firm net adding of products is associated with positive TFP growth, while firm net dropping of products is accompanied by a productivity decline. In the model, these associations are due to the fact that firms receiving positive productivity shocks earn greater revenue per existing product and are therefore able to extend their product range while those receiving negative shocks contract.

Though our model serves as a useful guide for our empirical analysis, it is too stylized in its current form to provide an explanation for all of the facts that we uncover. Within firms, for example, we find that some pairs of products are more likely to be co-produced than others, and, surprisingly, that mergers and acquisitions account for a relatively small share of

overall product switching. While these facts transcend our basic setup, we describe how the model might be extended to incorporate them. More generally, these additional facts provide additional guidance for further research on firm dynamics.

The remainder of the paper is structured as follows. Sections 2 and 3 outline our theoretical framework and describe our dataset. Sections 4 and 5 report our main empirical findings and their consistency with our theoretical framework. Section 6 concludes with suggestions for future research.

2. Theoretical Framework

In this section we outline a simple model of multi-product firms and product switching that is a natural extension of standard models of industry dynamics by Jovanovic (1982), Hopenhayn (1992), Ericson and Pakes (1995), Melitz (2003), and Bernard, Redding and Schott (2007). Our goal is to introduce the simplest model necessary for useful data analysis, that is, one that captures the essence of a broad class of models featuring product selection. Toward that end, we make use of a number of simplifying assumptions, for example ruling out supply- or demand-driven complementarities across products. We return to a discussion of how the model might be generalized and discuss alternative potential approaches after presenting our main empirical findings. We note that a more detailed discussion of the model and a more formal analysis of its implications is available in a web-based technical appendix.

2.1. Endowments and Preferences

Labor is the sole factor of production and is assumed to be in inelastic supply L (which also indexes the size of the economy). The representative consumer's preferences are a constant elasticity of substitution (CES) function of the consumption of a continuum of products $i \in [0,1]$:

$$U = \left[\int_0^1 (a_i C_i)^{\nu} \, di\right]^{\frac{1}{\nu}}, \qquad 0 < \nu < 1, \tag{1}$$

where $a_i > 0$ is a demand parameter that allows the relative importance of products in utility to vary. Firms are assumed to produce differentiated varieties of products, so that C_i is a

consumption index, which also takes the CES form:

$$C_i = \left[\int_{\omega\in\Omega_i} \left(\lambda_i(\omega)\, c_i(\omega)\right)^{\rho} d\omega\right]^{\frac{1}{\rho}}, \quad P_i = \left[\int_{\omega\in\Omega_i} \left(\frac{p_i(\omega)}{\lambda_i(\omega)}\right)^{1-\sigma} d\omega\right]^{\frac{1}{1-\sigma}}, \quad 0<\rho<1, \quad (2)$$

where ω indexes firm varieties within products, Ω_i is the (endogenous) set of firm varieties produced within product i, and P_i is the price index dual to C_i for product i.

The demand parameter $\lambda_i(\omega) \geq 0$ determines the representative consumer's relative demand for the varieties of different firms within each product.[7] Although not central to our results, we make the natural assumption that the elasticity of substitution across varieties within products is greater than the elasticity of substitution across products: $\sigma = \frac{1}{1-\rho} > \kappa = \frac{1}{1-\nu} > 1$. Similarly, we assume for simplicity that the elasticity of substitution across varieties within products, $\sigma = \frac{1}{1-\rho}$, is the same for all products.

2.2. Production Technology

Firms from a competitive fringe may enter by incurring a sunk entry cost of $f_e > 0$ units of labor. Incurring the sunk entry cost creates a firm brand and a blueprint for one horizontally differentiated variety of every product. Only once the sunk cost has been incurred does the firm observe its initial productivity, $\varphi \in [\underline{\varphi}, \bar{\varphi}]$, and consumer tastes for the characteristics embodied in its blueprint for every product, $\lambda_i \in [\underline{\lambda}, \bar{\lambda}]$. Productivity φ is firm-specific but is common across products within firms, whereas consumer tastes λ_i are firm-product specific and are therefore idiosyncratic to a particular product made by a particular firm.[8]

Productivity and consumer tastes evolve stochastically over time and we choose a specification for their evolution that is both tractable and sufficiently general to match key features of the firm-product data.[9] Upon entry, productivity and consumer tastes are drawn from the con-

[7] One interpretation of the parameter $\lambda_i(\omega)$ is product quality, though this parameter also captures other more subjective characteristics of a firm's variety that influence consumer tastes.

[8] With CES demand and monopolistic competition, differences in productivity across firms have identical effects on equilibrium revenue and profits to differences in consumer tastes. While we have modelled the component of profitability that is firm-product specific (λ_i) as consumer tastes, we could have equivalently introduced a productivity parameter that is firm-product specific. While consumer tastes are a plausible source of idiosyncratic shocks to firm-product profitability (see for example Foster et al. 2008), the important point for our analysis is that the profitability of a product for a firm has both a firm and a firm-product component.

[9] Bernard, Redding and Schott (2007) introduce a static model of multi-product firms without equilibrium product switching to examine the impact of trade liberalization on firms' product scope.

tinuous distributions $g_e(\varphi)$ and $z_{ei}(\lambda_i)$ respectively, with cumulative distributions $G_e(\varphi)$ and $Z_{ei}(\lambda_i)$. Once a firm observes its initial values of productivity and consumer tastes, it decides whether to produce or exit. If the firm exits, its production knowledge is lost, and the sunk cost must be incurred again in order for the firm to re-enter. If the firm enters, it faces a Poisson probability $\theta > 0$ of a shock to productivity φ, in which case a new value for productivity φ' is drawn from the continuous conditional distribution $g_c(\varphi'|\varphi)$, with cumulative distribution $Z_{ci}(\lambda_i'|\lambda_i)$. Similarly, the firm faces a Poisson probability $\varepsilon_i > 0$ of an idiosyncratic shock to consumer tastes for its variety λ_i, in which case a new value for consumer tastes λ_i' is drawn from a continuous conditional distribution $z_{ci}(\lambda_i'|\lambda_i)$, with cumulative distribution $Z_{ci}(\lambda_i'|\lambda_i)$.[10]

Consistent with the serial correlation in firm and firm-product shipments observed in our data, we assume that consumer tastes and productivity are positively serially correlated, which corresponds to the following assumption about their conditional distributions: $\partial Z_{ci}(\lambda_i'|\lambda_i)/\partial\lambda_i < 0$ and $\partial G_c(\varphi'|\varphi)/\partial\varphi < 0$ for $\lambda_i' \in [\underline{\lambda}, \bar{\lambda}]$ and $\varphi' \in [\underline{\varphi}, \bar{\varphi}]$. Thus the probability of drawing a new value for consumer tastes (productivity) less than λ_i' (φ') is decreasing in the existing value of consumer tastes (productivity). To make use of law of large numbers results, we assume that the distributions of consumer tastes and productivity are independent across firms. For the same reason, we also assume that the distributions of consumer tastes are independent across products and that the distributions of consumer tastes and firm productivity are independent of one another.[11] Firms also face a Poisson probability of death $\delta > 0$ due to *force majeure* events unrelated to profitability.

The production technology takes the following form. There is a fixed corporate headquarters cost of $f_h > 0$ units of labor, which the firm must incur irrespective of the number of products that it chooses to produce, and a fixed production cost of $f_{pi} > 0$ units of labor for each product i that is produced. In addition, there is a constant marginal cost for each product,

[10] While the model could be extended to allow firms to make endogenous investments in improving productivity and enhancing consumer tastes, these extensions are not central to the model's key predictions, which are driven by selection, and so we do not not pursue these extensions here.

[11] While the consumer taste and firm productivity distributions are independent of one another, there is interdependence in a firm's profitability across products, because firm productivity is common across products. We discuss extending the model to introduce other forms of interdependence in Section 5 below.

which depends upon the firm's productivity. Total labor employed by a firm with productivity φ is thus:

$$l(\varphi) = f_h + \int_0^1 I_i \left[f_{pi} + \frac{q_i(\varphi, \lambda_i)}{\varphi} \right] di, \tag{3}$$

where I_i is an indicator variable which equals one if a firm produces product i and zero otherwise, and $q(\varphi, \lambda_i)$ denotes output of product i by a firm with productivity φ and demand parameter λ_i.

2.3. Equilibrium Entry and Production Decisions

The key economic decisions of a firm in the model are whether to enter or exit and in which product markets to participate. We begin by considering the decision of whether to participate in a product market. As there is a fixed production cost for each product, there exists a **zero-profit consumer taste cutoff** $\lambda_i^*(\varphi)$ such that a firm with productivity φ will only produce product i if it draws a consumer taste greater than or equal to $\lambda_i^*(\varphi)$. The zero-profit consumer taste cutoff is defined as follows:

$$\pi_i(\varphi, \lambda_i^*(\varphi)) = \frac{R_i (\rho P_i \varphi \lambda_i^*(\varphi))^{\sigma-1}}{\sigma} - f_{pi} = 0, \tag{4}$$

where $\pi_i(\varphi, \lambda_i)$ denotes equilibrium profits from a variety of product i with consumer taste λ_i and firm productivity φ.

From equation (4), the higher a firm's productivity φ, the lower the zero-profit cutoff for consumer tastes $\lambda_i^*(\varphi)$, and so the greater the probability of having a value for consumer tastes sufficiently high to profitably produce the product. With a continuum of products and independent distributions for consumer tastes, the law of large numbers implies that the fraction of products produced by a firm equals the sum of its probabilities of producing each product. Therefore, as the probability of producing each product is increasing in firm productivity, a key implication of the model is that a firm's product range is increasing in its productivity.

We now consider a firm's decision of whether to enter or exit. With a continuum of products and independent distributions for consumer tastes, a firm's expected profits across the

continuum of products equals the sum of its expected profits from each product minus the fixed headquarters costs:

$$\pi(\varphi) = \int_0^1 \left[\int_{\lambda_i^*(\varphi)}^{\bar{\lambda}} \pi_i(\varphi, \lambda_i) \gamma_{zi}(\lambda_i) d\lambda_i \right] di - f_h. \tag{5}$$

where $\gamma_{zi}(\lambda_i)$ is the stationary distribution for consumer tastes, which as discussed further in the web-based technical appendix is endogenously determined as a function of the entry and conditional distributions, $z_{ei}(\lambda_i)$ and $z_{ci}(\lambda_i)$ respectively.

Although consumer tastes for a firm's variety of a product are stochastic, the law of large numbers implies that all firms with the same productivity experience the same flow of total profits across the continuum of products in equation (5). Stochastic shocks to consumer tastes generate fluctuations in the profitability of individual products, which lead them to be added and dropped over time. However, these fluctuations in the profitability of individual products average out at the level of the firm, so that the evolution of total firm profits over time is determined solely by stochastic shocks to firm productivity.

The value of a firm with productivity φ equals the flow of current profits plus the expected value of capital gains or losses as a result of a stochastic productivity shock, discounted by the probability of firm exit:

$$v(\varphi) = \begin{cases} \frac{\pi(\varphi) + \theta\left[\int_{\varphi^*}^{\bar{\varphi}} [v(\varphi') - v(\varphi)] g_c(\varphi'|\varphi) d\varphi'\right]}{\delta + \theta G_c(\varphi^*|\varphi)} & \text{for } \varphi \geq \varphi^* \\ 0 & \text{otherwise} \end{cases}, \tag{6}$$

where the denominator on the right-hand side of equation (6) is the probability of firm exit, which equals the exogenous probability of firm death δ plus the endogenous probability of experiencing a productivity shock that induces exit $\theta G_c(\varphi^*|\varphi)$.

The presence of a continuum of products in the model implies that each firm draws a value for consumer tastes above the zero-profit cutoff $\lambda_i^*(\varphi)$ in a positive measure of products. However, for entry to be profitable, the value of the current profits across this positive measure of products plus the expected value of capital gains or losses must exceed the fixed headquarters costs. As total firm profits across the continuum of products $\pi(\varphi)$ are increasing in φ, and the probability of experiencing a productivity shock that induces exit $G_c(\varphi^*|\varphi)$ is decreasing in

φ, the value of the firm in equation (6) is increasing in its productivity. Therefore, there is a zero-value cutoff productivity φ^* below which firms exit, which is defined by $v(\varphi^*) = 0$.

In an equilibrium with positive entry, the expected value of entry must equal the sunk entry cost, which requires the following free entry condition to hold:

$$V = [1 - G_e(\varphi^*)]\,\bar{v} = f_e, \qquad \bar{v} \equiv \int_{\varphi^*}^{\bar{\varphi}} v(\varphi) \left(\frac{g_e(\varphi)}{1 - G_e(\varphi^*)} \right) d\varphi, \tag{7}$$

where $[1 - G_e(\varphi^*)]$ is the *ex ante* probability of drawing a productivity above the zero-value cutoff φ^* upon entry, $g_e(\varphi)$ is the probability of drawing productivity φ upon entry, and $v(\varphi)$ is the solution to the Bellman equation (6).

General equilibrium is referenced by the following six variables and functions: the zero-value cutoff productivity below which firms exit φ^*, the zero-profit cutoff consumer taste for a firm with the zero-value cutoff productivity $\lambda_i^*(\varphi^*)$, the endogenous stationary distribution for firm productivity $\gamma_g(\varphi)$, the endogenous stationary distribution for consumer tastes $\gamma_{zi}(\lambda_i)$, the price index for each product P_i, and aggregate revenue for each product R_i. As shown in the web-based technical appendix, these six variables and functions are determined by the following equilibrium conditions: consumer and producer optimization, goods and labor market clearing, free entry, the zero-profit cutoff condition for consumer tastes for each product, the equality of the mass of new entrants and the mass of existing firms, the equality of the outflow and inflow of firms from each value of consumer tastes, and finally the equality of the outflow and inflow of firms from each value of productivity.

The general equilibrium of the model features both steady-state product switching and steady-state firm entry and exit. Each period a measure of new firms incur the sunk entry cost, and those with productivity draws above the zero-value cutoff enter, while those with productivity draws below the zero-value cutoff exit. A surviving firm with unchanged productivity produces a constant range of products, but idiosyncratic shocks to consumer tastes for individual products induce surviving firms to drop a measure of the products previously produced and add an equal measure of products not previously produced. As stochastic shocks to a surviving firm's productivity occur, the range of products produced expands with an increase in productivity and contracts with a decrease in productivity. Firms exit endogenously when

their productivity falls below the zero-value cutoff or exit exogenously when death occurs as a result of *force majeure* considerations.

As discussed further below, these and other features of the model are used to guide our empirical analysis. Before beginning that analysis, we describe our data.

3. Data Description

As part of its quinquennial Census of Manufactures (CMF), the U.S. Census Bureau (Census) collects information on the set of Standard Industrial Classification (SIC) categories produced by U.S. manufacturing establishments (i.e., "plants"). This information is obtained from questionnaires plants are required to fill out by law under Title 13 of the United States Code. Each questionnaire has two parts. The first is common to all establishments and solicits general information about their operation, including their overall shipments (i.e., "output"), use of inputs (capital, production and non-production workers and materials) and wagebills. We use this information to examine differences between single- and multi-product firms, and, along with industry price deflators provided by Bartelsman et al. (2000), to compute establishments' real labor productivity and total-factor productivity. The data appendix at the end of the paper provides further information on variable definitions.

The second part of each questionnaire varies depending upon the industry in which the establishment operates. It lists the set of products establishments in the industry typically produce as well as a verbal description of each product.[12] Establishments are instructed to record their total shipments of each product.[13] In the event that an establishment also ships products not listed on the form, the questionnaire provides space for them to record any shipments in additional product codes. Establishments are assigned to industries according to information collected from previous censuses as well as other Census surveys. Very large plants with sub-

[12] Questionnaires also collect information on establishments' "other" activities, such as "tasks performed for others using others' materials", that cannot be associated with a particular manufacturing product. We exclude these categories from our analysis. Questionnaires used for the 1997 census are available at www.census.gov.

[13] We note that the census does not collect information on firms' input use by product, and also that output deflators for five-digit SIC products are not available. As a result, measurement of establishments' use of inputs or productivity within individual products is not possible.

stantial activity in a number of industries may receive more than one form. Very small plants, referred to as "administrative records", are not required to report output at the product level. These establishments represent a very small share of overall U.S. manufacturing output and are typically ignored in U.S. microdata research; we drop them here as well.

We analyze product switching in the 1987 through 1997 censuses. We use a five-digit SIC category as our definition of a product and refer to two- and four-digit SIC categories as "sectors" and "industries", respectively.[14] As described in Census (1989), manufacturing encompasses 20 sectors, 455 industries and 1440 products. As product-mix decisions are made at the level of the firm, we aggregate plants to firms to create a firm by product by census year dataset.[15] Using this dataset we track the products that firms add and drop across census years.[16] Given that a considerable body of research already examines firm creation and destruction, we focus on the features of product switching by surviving firms highlighted by our model. In particular, we do not treat exiting firms as those who drop all their products, or entering firms as whose who add all their products. For convenience, we often refer to firms that produce multiple products as "MP firms" and firms that produce a single product as "SP firms".

Table 1 provides a sense of the relative level of detail between sectors, industries and products. It lists the products within the "Nonferrous Wiredrawing and Insulating" industry (SIC 3357), which is one industry inside the "Primary Metal Industries" (SIC 33) sector. The thirteen products in this industry range from copper wire (SIC 33571) to fiber optic cable (SIC

[14] Output at the five-digit SIC level is the most disaggregate data available for all plants. While CMF questionnaires solicit information at the seven-digit SIC level, establishments also surveyed for other programs (e.g., Census' Current Industry Reports) are permitted to report information at the five-digit level to alleviate their reporting burden.

[15] Firm identifiers are derived from firms' legal identities, and firms can consist of one or many establishments. Census uses an annual Company Organization Survey both to determine how new firms are organized and to keep track of changes in incumbent firms' ownership structure over time, e.g., the buying and selling of plants, the creation of new plants or the closing of existing plants.

[16] SIC categories undergo minor revisions in each census year but experienced a major revision in 1987. Census uses an internally generated concordance to map product codes collected in censuses after 1972 to the 1987 revision. We focus on the 1987 to 1997 censuses because they are less sensitive to this concordance and exhibit high product-code consistency over time. To be conservative, we drop the roughly 1 percent of five-digit codes (representing roughly 5 percent of total value) that do not appear in all three censuses. We note that our findings are not sensitive to this procedure, and that we find (but do not report) similar results for other sample periods, e.g., the 1972 to 1982 censuses, where the concordance of collected and 1987-revision product codes is less precise.

33579). Though these products share a grossly similar end use, they can differ substantially in terms of the materials and technologies required to manufacture them.

As indicated in Table 2, the typical two-digit sector has 24 four-digit industries and 76 five-digit products. The number of products per sector ranges from a low of 12 in Leather (SIC 31) to a high of 178 in Industrial Machinery (SIC 35). Similarly, the average number of products per industry within sectors ranges from a low of 1.1 in Leather to a high of 5.1 in Printing and Publishing (SIC 27).[17] Nonferrous Wiredrawing and Insulating (SIC 3357), the industry highlighted in Table 1, is just one of 26 Primary Metal Industries, and its products represent 14 percent (13/90) of the total number of products in that sector.

Products vary substantially in terms of how they are produced both within and across sectors. This variation is documented in the last four columns of Table 2, which summarizes products' 1997 capital and skill intensity by sector.[18] Mean capital intensity varies from a low of $52,000 per worker in Leather (SIC 31) to a high of $878,000 per worker in Petroleum (SIC 29), while skill intensity ranges from a low of 0.18 in Textiles (SIC 22) to a high of 0.48 in Printing and Publishing (SIC 27). Variation in products' input intensities within sectors is demonstrated by the standard deviations reported in columns six and eight. Implicit coefficients of variation for capital intensity are relatively high for Textiles (SIC 22), Rubber (SIC 30) and Electronics (SIC 36). For skill intensity, it is highest for Lumber (SIC 24).

We interpret the Standard Industrial Classification (SIC) categories used to record U.S. manufacturing output as discrete partitions of the model's continuum of products, which become coarser as one increases the level of aggregation. With this interpretation, the model provides a natural explanation for the coexistence of single- and multi-product firms. We think of firms producing a single product as those whose range of products falls within a single five-digit category. MP firms, on the other hand, are those whose product range is wide enough to span

[17] There is a substantial amount of variation in the precision of the industry and product classification. For example, Passenger Cars (SIC 37111) and Combat Vehicles (SIC 37114) are examples of products in the Motor Vehicle industry (SIC 3711), while Textbook Binding and Printing (SIC 27323) and Religious Books, Binding and Printing (SIC 27323) are examples of products in the Book Printing industry (SIC 2732).

[18] As noted above, the CMF does not collect information on input usage (or wages) by product. As a result, we measure a product's capital and skill intensity as the shipment-weighted average of all plants producing it.

several five-digit SIC categories.

Table 3 reports an average breakdown of SP and MP firms across the 1987 to 1997 census years in our sample, and also reports the average number of products, industries and sectors MP firms produce. As indicated in the table, MP firms dominate: though they represent a minority of firms (39 percent), they account for a strong majority (87 percent) of shipments. Multi-industry and multi-sector firms are similarly influential, responsible for 28 and 10 percent of firms but 81 and 66 percent of output, respectively. The final column of Table 3 reveals that the average MP firm produces 3.5 products, that the average multi-industry firm manufactures in 2.8 industries and that the average multi-sector firm is present in 2.3 sectors.[19]

4. Empirical Evidence

Our model highlights a number of features of product switching that operate at the level of firms, products and firm-products. We organize our empirical investigation of product switching in this section according to these levels of analysis. We note that the formal derivation of the model's implications is available in a web-based technical appendix.

4.1. Firm-Level Evidence

In the model, firms with higher productivity produce a wider range of products than firms with lower productivity because their higher revenues per product allow them to cover the fixed costs of a larger measure of products. Idiosyncratic shocks to firm-product profitability drive steady-state adding and dropping of products within firms, while idiosyncratic shocks to firm productivity induce changes in the measure of goods firms produce.

4.1.1. The Relative Productivity of Multiple-Product Firms

Table 4 compares the characteristics of single- and multi-product firms in the 1997 census, though we note that results are similar in previous census years. The table reports the results of OLS regressions of the natural log of the noted firm characteristic on a dummy variable

[19] On average across the census years 1987 to 1997, the share of MP firms with a single plant is 84 percent compared to a share of 93 percent for all firms. Therefore, MP firms are more likely to operate several production facilities than SP firms, but multiple products are frequently produced within the same production facility.

equal to unity if the firm produces multiple products as well as main-industry fixed effects. As indicated in the table, MP firms are larger than SP firms in the same industry in terms of both shipments (0.66 log points) and employment (0.58 log points). We also find that MP firms have higher output per worker and higher total factor productivity (TFP) than SP firms in the same industry.[20]

Similar differences are found with respect to firms producing in multiple industries and in multiple sectors, except for the TFP differential between single- and multiple-sector firms. That difference is statistically indistinguishable from zero, perhaps due to the difficulties of measuring productivity in firms with disparate products that span two-digit sectors.[21] All remaining differences displayed in the table are statistically significant at the 1 percent level.

4.1.2. Product Switching Within Firms

We examine product switching by dividing surviving firms into four exhaustive and mutually exclusive groups based on the manner in which they alter their mix of products between census years. Possible actions are: (1) *None* - the firm does not change its mix of products; (2) *Drop* - the firm only drops products; (3) *Add* - the firm only adds products; and (4) *Both* - the firm both adds and drops products, i.e., "churns" products.

Table 5 reports firm activity across these dimensions for the pooled 1987 to 1997 censuses. Cells in the top panel of the table report the average percent of firms reporting each activity across five-year census intervals, while cells in the bottom panel report percentages weighted by firm output. Though the figures reported in the table correspond to the probabilities of product switching conditional on firm survival, it is straightforward to evaluate the unconditional probabilities of product switching for all firms by multiplying the figures in the table by

[20] We measure firm TFP as the shipment-weighted average TFP of its plants. We measure a plant's TFP in a given census year relative to other plants in its main industry in percentage terms using the multi-factor superlative index number of Caves et al. (1982). This index accounts for plants' use of capital, production workers, non-production workers and materials. Plant shipments, capital and materials are deflated according to the four-digit SIC deflator of its major industry using deflators provided by Bartelsman et al. (2000). Wages are deflated by the U.S. consumer price index. A plant's main industry is the four-digit SIC in which it has the largest value of shipments.

[21] Our model implies that measuring the total factor productivity of multiple-product firms is problematic if, as is the case here, separate data are on output, prices and inputs at the firm-product level are unavailable. See also Bernard, Redding and Schott (2005) and De Loecker (2005).

the average probability of firm survival, which is roughly two-thirds. The five columns in each panel report results for all firms, MP firms, firms that export, firms whose shipments are above the 75th percentile ("large firms") and firms with more than one manufacturing plant.

As indicated in the upper panel of the table, an average of 54 percent of surviving firms alter their mix of products every five years, 15 percent by dropping at least one product, 14 percent by adding at least one product and 25 percent by both adding and dropping at least one product. Comparing the results for all firms in the first column with those for MP firms in the second column, we find implicitly that SP firms are more likely to leave their product mix unchanged than MP firms. From the third column of the table, we find that exporters are more likely to change their product mix than non-exporters. Finally, from the remaining columns of the table, we see that large firms and multi-plant firms also have above-average rates of product switching.[22]

The frequency and pervasiveness of product switching displayed in Table 5 is consistent with our model. The bottom panel of the table reveals that large firms are relatively more likely to add and drop products than small firms. This behavior is also understandable in light of our theoretical framework, as more productive firms have product ranges wide enough to span five-digit SIC categories, rendering them more likely to add and drop products. Since more productive firms in the model also produce more of each product and have larger total output, firms that switch products are likely to account for a larger share of output than of the number of firms. As indicated in the table, an average of 89 percent of all manufacturing output is produced by firms that change their mix of products across census years. Firms that both add and drop products account for the largest share of output, at 68 percent.

[22] Results for SP firms, non-exporters, "small" firms and single-plant firms are available upon request. We have also performed an alternate decomposition of activity according to whether firms (1) do not change their product mix; (2) change their mix but do not net add or drop any products; (3) change their product mix and net add products; and (4) change their product mix and net drop products. On average across all firms and census years, these behaviors occur 46, 12, 22 and 20 percent of the time.

4.1.3. Product Switching and Firm Characteristics

We examine the relationship between product switching and firm outcomes via OLS regressions of log changes in firm characteristics between census years on dummy variables capturing contemporaneous product-switching behavior,

$$\Delta Z_{jt} = \alpha_{mt} + \alpha_m + \beta_1 NetDrop_{jt} + \beta_2 NetAdd_{jt} + \varepsilon_{jt}. \tag{8}$$

where ΔZ_{jt} represents the log difference in a firm outcome between Census years $t-5$ and t; α_{mt} represents a full set of product mix by year fixed effects; $NetDrop$ is a dummy variable that is equal to one if a firm reduces its net number of products and zero otherwise; and $NetAdd$ is a dummy variable that is equal to one if a firm increases its net number of products and zero otherwise.[23] The firm characteristics we consider are real output, employment, real labor productivity and total factor productivity[24]. The regression results, reported in Table 6, include all surviving firms between the 1987 to 1992 and 1992 to 1997 censuses. The regression coefficients therefore capture the correlation between changes in the net number of products and changes in firm characteristics conditional on firm survival. Each row of the table reports results for a different firm-outcome regression. Standard errors adjusted for clustering at the industry level are reported in parentheses below coefficients. The number of firm-year observations included in each regression as well as each regression's R^2 are reported in the final two columns of the table.

As indicated in the table, we find that product switching is related to changes in firm characteristics in the way suggested by the model. We find that net product adding is associated with an increase in firm size (whether measured by output or employment) as well as produc-

[23] The left out category encompasses firms that undertake no product switching or, if they do switch products, experience no net change in the number of products they produce. Similar results are obtained if an additional dummy variable is included for firms that engage in product switching but experience no net change in products. Though our regression focuses on net adding and dropping because these relate most closely to the predictions of the model, we note that an analogous specification using the Add, Drop and Both measures defined above also reveals statistically significant correlations between product switching and changes in measured firm characteristics.

[24] Results for nominal output and nominal output per worker are similar. Given the unavailability of product-level price indexes, firms' product-level shipments are deflated by their corresponding industry-level deflators. The inclusion of product mix by year fixed effects in the regression helps to alleviate concerns about product-year variation in prices.

tivity (whether measured by labor or total factor productivity). Similarly, we find that net product dropping is associated with a decrease in firm size and TFP. Although the correlation between net product dropping and labor productivity is positive rather than negative, the estimated coefficient is roughly an order of magnitude smaller than those for the other variables. As noted above, the structure of our model implies that measuring the productivity of multi-product firms is problematic when data on inputs and price deflators are unavailable at the firm-product level.

While the regression results in Table 6 establish that product switching is accompanied by changes in observed firm characteristics, we emphasize that they are correlations capturing an equilibrium relationship between endogenous variables. As product choice is endogenous, the regression coefficients capture both the non-random decision to change the net number of products and the impact of this decision on observed firm characteristics.

4.1.4. Potential Product-Category Mismeasurement

Census devotes considerable resources to the accurate collection and verification of establishments' product-shipment data. As noted above, forms are designed to minimize measurement error by being tailored to the industry in which establishments operate, by listing the SIC categories (and descriptions) that establishments in the industry commonly produce, and by offering establishments space to record output in unlisted categories. After forms are collected, Census verifies the consistency of current responses with past responses and re-contacts establishments whose data appears erroneous. Nevertheless, our analysis of product switching is susceptible to establishments' inaccurate transcription of SIC codes.

We believe our results to be robust to product-category mismeasurement for several reasons. First, we note that our use of five-digit SIC categories to define products requires only that firms correctly record the first five digits of the seven-digit SIC categories listed on the Census questionnaire. Second, we find little evidence of spurious product switching in the data: less than two percent of firms across the 1987 to 1997 censuses, for example, are observed producing, not producing, and again producing the same product. Third, the results reported in Table 6

indicate correlations between product-switching behavior and separately-recorded measures of firm characteristics such as size and input usage that are systematic and consistent with our model. The consistency of these and other empirical results with the predictions of the model is hard to square with simple explanations of mismeasurement based on classical measurement error.

Fourth, we note that we observe similar switching behavior with respect to even more easily identified four-digit SIC industries and two-digit SIC sectors. Table 7 compares firms' extensive-margin adjustments for products (column 1; reproduced from Table 5), industries (column two) and sectors (column three) – using the same typology of activities as in Table 5. The first row of the table records the average share of firms making no adjustments between census years. Unsurprisingly, product switching (54 percent) is more likely than industry switching (41 percent), and industry switching is more prevalent than sector switching (16 percent). Even so, product adding induces an average of 27 percent of firms to enter at least one new industry and 9 percent of firms to break into at least one new sector every five years. To the extent that adding industries and sectors requires adopting unfamiliar production and distribution technologies, these findings also suggest that firms extensive-margin adjustments involve considerable changes in the nature and scope of firms.

As a final check, we also examine how product switching varies depending on the main two-digit manufacturing sector of a firm. We find that the average percent of firms that alter their mix of products every five years varies from a low of 33 percent in Stone & Concrete (SIC 32) to a high of 71 percent in Printing & Publishing (SIC 27), as reported in Table 16. Product switching therefore appears to be a pervasive feature of the U.S. manufacturing sector that is not driven by behavior in a few influential sectors.

4.2. Product-Level Evidence

In the model, there is a positive correlation between products' add and drop rates: while some firms not producing a product receive a positive demand shock and therefore add it, some of the incumbent producers receive negative demand shocks and so drop it. Variation in

product add and drop rates is governed by the probability of receiving a shock. "Turbulent" product markets, where idiosyncratic shocks are more likely, exhibit more frequent adding and dropping, other things equal, than "stable" products where shocks are less prevalent. A related feature of product switching at the product level concerns gross versus net changes in product output. As idiosyncratic shocks lead different sets of firms to add and drop the same product simultaneously, the model has gross changes in product output dominating net changes.

4.2.1. Product Add Versus Drop Rates

Figure 1 displays the mean rate at which five-digit SIC products are added and dropped by U.S. manufacturing firms across the 1987 to 1997 censuses. A product's add rate in year t is computed as the number of firms adding the product between census years $t-5$ and t divided by the average number of firms producing the product in both years. Drop rates are computed analogously. As shown in the figure, there is a clear positive correlation between the rates at which products are added and dropped. This correlation is statistically significant at conventional levels.[25]

The positive correlation between the rates at which U.S. manufacturing products are added and dropped indicates that the extensive-margin adjustments we observe in the data cannot be explained solely in terms of a net reallocation of economic activity from one group of products to another. Such a net reallocation would imply a negative correlation between the rates of product adding and dropping, as growing products are frequently added and infrequently dropped, and declining products are frequently dropped and infrequently added. Although the fact that add and drop rates do not lie perfectly along a 45 degree line indicates that there is some net transfer of output across products in the data, other forces are also clearly at work.[26]

We note that while the positive correlation in Figure 1 is hard to reconcile with pure net

[25] Existing research on plant creation and destruction finds a positive correlation between firm entry and exit rates. See for example Dunne et al. (1988) for the United States and Disney et al. (2003) for the U.K..

[26] In the model, a turbulent product with a high probability of idiosyncratic shocks to demand not only has high rates of product adding and product dropping but also displays a high volatility of shipments at firms that continue to produce it. Consistent with this, we find a positive correlation in the data between a product's rate of adding or dropping and its mean standard deviation of log shipments over time at firms that continue to produce the product.

reallocation *across* products, it does not rule out unobserved net reallocation *within* products. Indeed, in the model, net reallocation within products occurs because some firms add a product as other firms drop it.

4.2.2. Product Switching and Aggregate Output

Our model points to product switching as a new dimension of resource reallocation that complements the more widely-studied margin of firm entry and exit. To assess the relative importance of this new dimension at the product and aggregate level, we decompose a product's output according to the type of firm producing it. In our first decomposition, we look backward in time to divide product output in year t according to firms that produce the product in both t and $t-5$ ("incumbents"), surviving firms that do not produce the product in $t-5$ but do produce it in t ("adders"), and firms that do not exist in $t-5$ but produce the product in t ("entering firms"),

$$Y_{tp} = \sum_{j \in B_{tp}} Y_{tpj} + \sum_{j \in A_{tp}} Y_{tpj} + \sum_{j \in N_{tp}} Y_{tpj}, \tag{9}$$

where p indexes products; j denotes firms; and B_{tp}, A_{tp} and N_{tp} represent the set of incumbents, adders and entering firms, respectively.

Our second decomposition is forward-looking and divides a product's output in year t according to firms that produce the product in both t and $t+5$ ("incumbents"), surviving firms that produce the product in t but not in $t+5$ ("droppers"), and firms that produce the product in t but die between t and $t+5$ ("exiting firms"),

$$Y_{tp} = \sum_{j \in B_{tp}} Y_{tpj} + \sum_{j \in D_{tp}} Y_{tpj} + \sum_{j \in X_{tp}} Y_{tpj}, \tag{10}$$

where D_{tp} and X_{tp} denote the sets of dropping and exiting firms, respectively.

The decompositions in equations (9) and (10) have a couple of attractive features for our analysis. First, because they are based on the nominal value of output in year t, they do not require product-level price deflators. Second, they can be converted into percentage decompositions for each product by dividing through by Y_{pt}. As a result, they do not require comparisons of output value across products and so avoid the problems associated with such comparisons.

The upper panel of Table 8 reports the mean product-value decompositions in percentage terms across all products. Each row of the panel reports the average decomposition for a particular census year, with the first three columns looking backward (adding versus entering) and the final three columns looking forward (dropping versus exiting). In both cases we find that roughly two-thirds of the average product's output is produced by incumbents. The remaining output is more or less evenly split between firms adding or dropping the product and entering or exiting firms. In 1992, the only year of the sample for which both decompositions can be performed, adders and entrants are responsible for an average of 14 and 19 percent of products' output, respectively, while droppers and exiters account for 15 and 18 percent, respectively.

The lower panel of Table 8 reports the result of a similar decomposition for the share of firms producing a product in a census year. While incumbents again make the greatest single contribution, their average share of firms, at 40 to 45 percent, is lower than their average share of output. Of the remaining 55 to 60 percent of producers, 29 to 37 percent are entering or exiting firms and 23 to 27 are adders or droppers. Table 16 reports the results of these decompositions by two-digit sector. While there is some variation across sectors, we find substantial contributions of roughly equal magnitude from adders and droppers and entering and exiting firms in each two-digit manufacturing sector. As product entry/exit occurs as a result of either the addition/deletion of products within surviving firms or the entry/exit of firms, the results of these decompositions show that these two dimensions of product entry/exit are of roughly equal importance, whether measured in terms of the frequency of product entry/exit or the value of output.

The breakdowns reported in Table 8 also highlight the fact that gross changes in product output are substantially larger than the associated net changes, an "excess reallocation" that is similar in spirit to the one found in job creation and destruction by Davis and Haltiwanger (1992). Indeed, comparison of the forward-looking 1992 decomposition with the backward-looking 1997 decomposition reveals that 15 percent of the average product's 1992 output is accounted for by firms that subsequently drop the product, while 15 percent of 1997 output is due to firms that just added it. Over the same period, the change in the average share of

output represented by incumbents was just 3 percent (from 67 to 70 percent).[27]

4.3. Firm-Product Level Evidence

In the model, a firm's profitability in a particular product is the result of an interaction between firm-wide productivity and consumers' taste for particular firm-products. As a result, firms adding products are likely to have higher values of firm productivity than firms whose product range remains constant. Likewise, because consumer tastes are serially correlated and so are positively correlated with a firm-product shipments and the length of time for which a firm has produced a product, firms are more likely to drop products that are small (scale dependence) or relatively new to the firm (age dependence). Finally, variation in consumer taste across products within firms results in differences in the size of shipments across products within firms.

4.3.1. Product Adding

Firms' product adding decisions are systematically related to their productivity in existing products in the way suggested by the model. We find a positive association between initial firm productivity and subsequent product adding among firms producing the same initial mix of products. Table 9 reports the results of OLS regressions of a dummy variable indicating product adding by either SP (right panel) or MP firms (left panel) between 1992 and 1997 on firm productivity in 1992,

$$Add_{jt} = \alpha_m + \beta_1 Productivity_{jt} + \varepsilon_{jt}, \tag{11}$$

where α_m represents a set of product-mix fixed effects and the productivity of firm j is measured in terms of either labor productivity or TFP. We employ a linear probability model so that product-mix fixed effects, which allow for a comparison of behavior among firms producing the same initial set of products, can be included in the regression. Given that we estimate

[27] We find a similar dominance of gross versus net product switching in a decomposition of real U.S. manufacturing growth that separates real output changes according to firm entry and exit, incumbents' product adding and dropping and incumbents' continuing-product growth and decline. A disadvantage of that decomposition relative to the one presented here is its reliance on the same industry-level price indexes to deflate the output of all products within the same industry.

the regression for a single cross-section, the product mix fixed effects control for the level and change of any product-mix-specific characteristic that influences the probability of adding a product between 1992 and 1997.[28]

As shown in the first and third columns of each panel of Table 9, subsequent product adding is positively and statistically significantly correlated with both initial TFP and initial labor productivity for both SP and MP firms. As noted in the second and fourth columns of each panel, this positive correlation remains when controls for firm size (i.e., employment) and age are included in the regression. These results are subject to the aforementioned caveats about the problems of measuring firm productivity when separate data on output, inputs and price deflators are not available at the product level. Nonetheless, they suggest that the productivity advantage of MP firms observed above in Table 4 is due at least in part to selection: SP firms who subsequently become MP firms are on average more productive than other SP firms.[29]

4.3.2. Product Dropping

We also find evidence of scale and age dependence in firms' decisions to drop products in line with the process of selection within firms emphasized by the model. Table 10 reports OLS regressions of a dummy indicating the dropping of one of a surviving firm's products between census years 1992 and 1997 on firms' 1992 relative product size ($Size_{jit}$) and relative product tenure ($Tenure_{jit}$) as well as both firm and product fixed effects:

$$Drop_{jpt} = \alpha_j + \alpha_p + \beta_1 \ln(Size_{jpt}) + \beta_2 \ln(Tenure_{jpt}) + \varepsilon_{jpt}, \tag{12}$$

where j and p index firms and products respectively and α_j and α_p represent firm and product fixed effects, respectively. The variables $Size_{jit}$ and $Tenure_{jit}$ are defined in terms of shipments and the length of time for which a firm has produced a product respectively. Both size and

[28] We find similar results for earlier census periods. The analogous specification when census periods are pooled involves including a full set of interactions between product mix fixed effects and time fixed effects. This specification also yields similar results.

[29] Our finding of a positive correlation between a firm's productivity given its existing set of products and its decision to add a new product is hard to reconcile with a model in which products are randomly assigned to firms (see for example Armenter and Koren 2008). Under random assignment, there would be no correlation between existing producer characteristics and the addition of a new product.

tenure are measured relative to their averages for the product via log differencing in each Census year. As a result, these variables control for differences across products in output and tenure, both at a point in time and over time. We note that we examine the model's firm-product predictions in the context of product dropping because construction of an analogous adding sample is impractical given the size of our dataset. Our sample consists of surviving firms and therefore the estimated coefficients capture the determinants of a firm's decision whether to drop products conditional on firm survival.

As our regression specification is estimated for a single cross-section of data based on the decision whether to drop a product between 1992 and 1997, the firm fixed effects control for any firm characteristic that is common across products and affects the decision whether to drop a product over this period (e.g., total firm shipments, the growth of total firm shipments, firm age, whether a firm is an exporter or enters/exits export markets, whether a firm has multiple plants). Similarly, the product fixed effects control for any product characteristic that is common across firms and influences the decision whether to drop a product (e.g., an aggregate change in relative demand or supply across products). The coefficients β_1 and β_2 are therefore identified solely from the variation in shipments and tenure that is idiosyncratic to individual pairs of firms and products.[30]

Results are reported with and without firm and firm-plus-product fixed effects. In all three cases, coefficient estimates indicate that firms are less likely to drop a product if their shipments and tenure are large relative to firms producing the same mix of products.[31] To the extent that relative firm-product size and tenure are positively correlated with firm-product productivity, the results in Table 10 suggest a systematic reallocation of economic resources

[30] We find similar results for earlier census periods. The analogous specification when census periods are pooled involves including a full set of interactions between firm and time fixed effects and between product and time fixed effects. This specification also yields similar results.

[31] In the model, consumer tastes follow a first-order Markov process, so that the probability of drawing a new value for consumer tastes depends only on the current value of consumer tastes. Furthermore, controlling for firm and product fixed effects, log firm-product shipments are proportional to the current value of consumer tastes. Therefore, as in much of the firm entry and exit literature, age or tenure should become insignificant in a specification that controls appropriately for scale. One natural explanation for the significance of firm-product tenure in such a specification is that consumer tastes follow a higher-order Markov process, and the model could be extended to allow for this possibility.

within firms towards higher-productivity activities. As a result, studies of industry dynamics that ignore firms' extensive margins likely underestimate the role of reallocation in both output and productivity growth.

4.3.3. Product Switching and Firm Output

We find that the process of reallocation within firms captured in the model is quantitatively important at the firm level as well as at the aggregate level. To illustrate this, we decompose the output of surviving firms in a given census year according to whether the products are continuously produced versus recently added or about to be dropped. These backward- and forward-looking firm-level decompositions are analogous to those used for products in equations (9) and (10), respectively, above. Here, however, there is no contribution from firm entry or exit because the decompositions are undertaken for surviving firms.[32] As shown in Table 11, we find that on average 26 and 31 percent of firm output in 1992 and 1997, respectively, is represented by products firms added within the previous five years. A comparable average share of firm output, 29 and 26 percent for 1987 and 1992, respectively, is accounted for by about-to-be-dropped products. These shares suggest that product switching exerts considerable influence on firm activity, and that gross changes in firm output are substantially larger than net changes.

Our model also highlights a distinction between a firm's extensive margin (number of products) and its intensive margin (output per product). We therefore decompose the cross-section variation in firm sizes into the contributions of these extensive and intensive margins. Here we find a smaller contribution from the extensive margin than in the results presented above because variation in only the *net* number of products is considered. Across firms within a given census year, we find that firms' intensive and extensive margins account for approximately 90 percent and 10 percent of the variation in firm size, respectively. These contributions are obtained from OLS regressions of firms' log average shipments per product and firms' log number

[32] We note that the product-level decompositions reported earlier are not simple averages of the firm-level decompositions reported here for additional reasons besides the focus on surviving firms. In particular, the weight of firms in the product-level decompositions varies substantially depending on their size, and the firm-level decompositions include a firm's output across all products.

of products on firms' log total shipments. Given that the latter must equal the sum of the former for each firm, the two coefficients from this regression capture the average contributions of the intensive and extensive margins to cross-sectional variation in firm size, respectively. Regressions performed separately in each census year from 1987 to 1997 yield similar results.

4.3.4. Distribution of Product Shipments Within Firms

To provide evidence on the product heterogeneity within firms featured in the model, Table 12 reports the average share of firm output represented by each of a firm's products, with products sorted from largest to smallest. To conform with Census disclosure requirements, we report these average shares for firms producing up to ten products. We note that firms producing ten or fewer products represent roughly 99 percent of firms and roughly half of U.S. manufacturing shipments in our sample. As shown in the table, the distribution of output across products is highly skewed, with the average share of firm output attributable to a firm's largest product declining from 80 percent for firms that produce two products to 46 percent for firms that produce 10 products.

A commonly used benchmark in the literature on firm size distributions is the Pareto distribution, which predicts a log linear regression relationship between the log rank of firm shipments and log firm shipments. To similarly assess the product-size distribution within firms producing a like number of products, we estimate an analogous regression of the log rank of firm-product size on the log of their share of firm shipments.[33] We estimate this regression separately for firms producing different numbers of products using the data on the average shares of products in firm output reported in Table 12. The fitted and actual values for firm-product rank and size in these regressions are displayed in Figure 2. As indicated in the figure, actual values lie above the regression line in the middle of the distribution and below the regression line in the tails, implying thinner tails than the Pareto distribution.[34] Therefore the heterogeneity across

[33] If the distribution of shipments, x, across products within firms shown in Table 12 is Pareto with minimum value k and shape parameter a, we have $\text{Prob}(x > x') = (k/x')^a$. Taking logarithms in this expression and rearranging terms yields the following relationships: $\log(Rank_p) = A - a\log(x_p) = B - a\log(Share_p)$, where $Rank_p$ is the rank of x_p, $Share_p = x_p/X$, and A, B and $X = \sum_p x_p$ are constants.

[34] Including a quadratic term in log product size in the regression, we find that a null hypothesis of linearity is strongly rejected at conventional levels of statistical significance. From a comparison of the tails across the

products within firms stressed in our model displays the same features as the heterogeneity across firms examined in the firm-size distribution literature (see, for example, Rossi-Hansberg and Wright 2007).

4.4. Alternate Explanations

Our empirical analysis of product adding and dropping thus far accords well with features of product switching highlighted by our model of endogenous product selection. Here, we discuss potential alternate explanations for the facts we uncover and the extent to which they receive support from the data.

Explanations of product switching fall into three broad categories according to whether they focus on factors that are specific to products, on factors that are specific to firms, or on factors that are idiosyncratic to firm-product pairings. The first category of explanations emphasizes forces that are product-specific but common to all firms, such as changes in relative demand (e.g., changing fashions) or relative supply (e.g., changing technology). Explanations of this form that involve a net reallocation of economic activity across products, e.g., from "cold" to "hot" products, are hard to reconcile with the positive correlation between products' add and drop rates observed in Figure 1.

A second class of explanations for product switching focuses on factors that are specific to firms but common to products. Positive shocks to a firm's productivity, for example, might increase the profitability of all products it could produce, thereby inducing the firm to add previously unprofitable products. This class of explanations, however, is hard to reconcile with the fact that firms simultaneously add and drop products across census years. Such switching suggests that any firm-specific shocks differentially affect its products, that is, that shocks are firm-product specific. A more fundamental challenge for both firm- and product-specific explanations of product switching is our finding that firm-product characteristics are influential determinants of product switching even after controlling separately for firm and product characteristics.

panels of Figure 2, the departures from a Pareto distribution increase with the number of products that firms produce.

The model developed in Section 2 falls into the third category of explanations, which concentrates on the role of firm-product attributes in influencing product switching. In our model, the interaction of idiosyncratic shocks to firm productivity and firm-product demand fosters both self-selection of firms and self-selection of products within firms. Klette and Kortum (2004) – hereafter KK – offer an alternate firm-product approach that emphasizes innovation.[35] In the KK model, products cycle across firms as they exchange technological dominance. While this model is consistent with some of the stylized facts we present (e.g., product switching across census years), it fails to capture others. In KK, for example, the firm-size distribution is determined entirely by variation in the extensive margin of the number of products firms produce. The regression decomposition discussed above, however, indicates that the intensive margin of output per product is quite influential in determining variation in firm size. The KK model also predicts a constant hazard rate – equal to the economy-wide rate of creative destruction – for firms' dropping of products. Here, however, we find that the probability that a firm drops a product is decreasing in firm-product shipments and the length of time for which the firm has produced the product. Though our data motivate the development of the selection model described above, we believe that extending innovation-based models to match our new stylized facts is another interesting avenue for further research.

5. Extending the Basic Selection Model

In this section we highlight several dimensions of the data that are less well captured by either our model or the alternative potential explanations discussed above, but which point to potentially fruitful lines of future theoretical and empirical research.

5.1. Product Co-Production

Our first set of additional results relates to the types of goods firms tend to produce together. Table 13 reports the average annual frequency, in thousands, with which firms co-produce products within and across sectors from 1987 to 1997. Dark shading indicates co-production that

[35] See also Lentz and Mortenson (2005), Luttmer (2007) and Chatterjee and Rossi-Hansberg (2008) for other innovation-based models of firm scope.

is significantly more frequent than expected based on the individual probabilities of producing each product, while light shading indicates significantly less co-production.[36] As shown in the table, the probability that a firm produces a product in the row sector conditional on production of a product in the column sector is relatively high within sectors as well as between sectors that appear related (e.g., Apparel and Textiles, or Electronics and Industrial Machinery). Furthermore, the matrix of data as a whole rejects the null hypothesis that the probability a firm produces a product is independent of the firm's other products (p-value < 0.01).

In developing our model, we assumed for simplicity that consumer tastes λ_i were independently distributed across products. As discussed in footnote 8, the consumer taste parameter λ_i has an equivalent interpretation as a component of firm productivity that is product-specific, and therefore the more general assumption is that the product-specific component of firm sales is independently distributed across products. Under this assumption, the only interdependence in firm sales across products arises from φ, which raises or reduces a firm's sales across all products proportionately. While this interdependence is consistent with our finding that more productive firms are more likely to add products, the assumption that consumer tastes are independently distributed across products sits awkwardly with the results reported in Table 13. As apparent from the table, some pairs of products are systematically co-produced within firms, while other pairs of products are systematically produced in separate firms.[37]

These empirical findings of patterns of co-production are suggestive of positive interactions (complementarities) in demand or production technology between some pairs of products and/or negative interactions in demand or production technology between other pairs of products. One way of extending the model to incorporate such relationships is to allow consumer

[36] We assess statistical significance by comparing the observed co-production frequencies to those that would be expected under a null hypothesis that the decisions to produce row and column product lines are independent. Under this null, the expected frequency with which a particular pair of major groups is co-produced follows an independent Poisson distribution. An individual cell's deviation from random co-production therefore follows a standard normal distribution, $\frac{o_{rc}-e_{rc}}{\sqrt{e_{rc}}} \sim N(0,1)$, where o_{rc} and e_{rc} are the observed and expected frequencies in row r and column c, respectively. Summing across cells, the statistic for testing whether the entire matrix of frequencies is generated by random co-production, $\sum_{r,c} \frac{(o_{rc}-e_{rc})^2}{e_{rc}}$, is distributed chi-squared (Cochran 1952).

[37] Similarly, the co-production findings in Table 13 sit awkwardly with the assumption in the innovation-based model of Klette and Kortum (2004) that the identity of the product to which a firm's innovation applies is drawn randomly from the set of potential products.

tastes (or the product-specific component of productivity) to be either positively or negatively correlated across products. For example, product characteristics that are highly valued in one product market (e.g., apparel) may be highly valued in another product market (e.g., textiles), or technical expertise in the production of one product (e.g., cars) may be relevant in the production of other products (e.g., motorcycles). While such an extension would bring the model closer to the data, this would come at the cost of making the model considerably less tractable. Nevertheless, achieving greater understanding of the sources of interdependence in demand or production technology across products within firms is an important area for future theoretical and empirical research.

5.2. Mergers & Acquisitions

Our second set of additional results relates to the manner by which firms add and drop products. There are a variety of ways in which firms can add a product: at existing facilities, at newly constructed plants, or by acquiring an existing plant from another firm. Similarly firms can drop products at continuing plants, by closing plants or by selling plants to another firm.

Surprisingly, we find that relatively few of the changes that firms make to their mix of products occur because of merger and acquisition (M&A) activity. Indeed, the vast majority of switches take place within firms' existing facilities. Table 14 reports the distribution of product adds (top panel) and drops (bottom panel) according to how they are accomplished. As indicated in the first column of each panel, roughly 85 percent of added and dropped products, respectively, are added and dropped by existing plants. The share of products added and dropped through M&A is small: less than 10 percent of both adds and drops involve plant acquisitions or divestitures whether by themselves or in combination with another mode of product switching. As shown in the second column of each panel, M&A activity is more important as a share of product value, indicating that the products added and dropped through plant acquisitions or divestitures are on average larger than those added and dropped through other modes of product switching. Columns three and four of each panel show that a similar pattern is observed for the share of firms that add and drop products. Less than 5 percent

of firms add and drop products through M&A or through a combination of modes involving M&A, but these firms account for a substantial share of the value of manufacturing output.

Though product switching is not synonymous with M&A activity, M&A activity is tightly linked with switching. Table 15 compares product and sector-switching according to whether or not firms concomitantly acquire or divest a plant. As indicated in the table, firms involved in an ownership change are relatively more likely to change their mix of products. An average of 94 percent of firms that engage in M&A activity also alter their mix of products, compared with an average of 53 percent for firms that do not participate in an acquisition or divestiture. For sector switching, the importance of M&A is even more stark: the analogous percentages are 67 and 15 percent.

Therefore, although plant acquisitions and divestitures account for a relatively small fraction of the products added and dropped, M&A activity is disproportionately likely to result in product switching. While our finding that product switching is concentrated at firms' existing plants motivates the model's abstraction from M&A and the creation of new plants, the introduction of these alternative modes of product switching is an interesting area for further research.

6. Conclusions

The extent to which resources are allocated to their best use is a primary concern of economics. Virtually all empirical research on reallocation as a source of industry output and productivity growth focuses on firm entry and exit or changes in the composition of output across firms. This paper identifies product switching as an important source of reallocation within firms and analyzes its determinants and consequences.

Guided by a natural extension of existing models of industry dynamics that allows firms to produce an endogenous range of products in response to evolving firm and product characteristics, we develop a body of evidence about this new dimension of firm behavior. Using a novel dataset that tracks U.S. manufacturing output at the level of five-digit products within firms, we find that firms add and drop products with surprising intensity and frequency. On

average, 54 percent of US manufacturing firms alter their mix of five-digit products every five years, and these adjustments lead an average of 41 percent of firms to enter new or exit existing four-digit industries, and 16 percent of firms to extend or contract their set of two-digit sectors. Overall, we find that the gross contributions of product adding and dropping to the evolution of aggregate manufacturing output are as large as the gross contributions of firm entry and exit.

We demonstrate that observed patterns of product switching are inconsistent with explanations based purely on net reallocation across products and are more generally hard to reconcile with explanations based on firm or product shocks alone. In contrast, we find support for the central features of our extended model of industry dynamics, which emphasizes selection within as well as across firms. In particular, the model accounts for the positive correlation across products between the rate of product adding and dropping and the age and scale dependence observed in the probability a product is dropped.

Though our basic framework is a good match for key features of the census data, additional empirical analysis reveals areas in which it might be extended. In the current version of the model, for example, the only source of dependency in profitability across a firm's products is the firm's overall productivity: higher firm productivity raises the profitability of all products. Empirical examination of firms' product mix, however, reveals that firms are more likely to co-manufacture products within the same industry, or within "linked" industries, e.g., lumber and furniture or electronics and instruments. An extended version of our model might incorporate demand- or supply-side complementarities that rationalize these links.

Analysis of how firms add and drop products reveals that the majority of product switching occurs within firms' existing plants. This finding motivates our model's abstraction from M&A and the creation of new plants. In the data, however, we find that find that product adding that does take place via merger or acquisition (M&A) is more likely to extend a firm's reach into new four- and two-digit SIC industries than product adding that takes place within existing plants. Extending the model to allow firms to endogenously choose between alternative modes of product switching is therefore an interesting area for further research.

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A Data Appendix

As noted in Section 3, our data come from the U.S. Census Bureau's quinquennial Census of Manufactures (CMF). At the establishment-year level we observe establishments' total value of shipments (which we also refer to as output), which is defined by Census as: "the value of the products shipped and services performed at the net selling value, f.o.b. plant to the customer; i.e., after discounts and allowances, and exclusive of freight charges and excise taxes." We also observe plants' use of production workers and their wagebill; their use of non-production workers and their wagebill; the value of materials used; and the total value of plant and equipment. All values in the CMF are nominal. To compute establishments' labor productivity and total factor productivity, we deflate nominal shipments, cost of materials and capital using the industry-level deflators from Bartelsman et al. (2000). For each establishment, we use the industry deflator that corresponds to its major four-digit inustry. We use the U.S. Consumer Price Index from www.bls.gov to deflate wages.

Table 16 summarizes several of our main findings by two-digit SIC sector. The table in which the main result can be found is noted at the top of each column. Column 1 reports the mean share of multi-product firms across the 1987 to 1997 censuses. Column 2 reports the mean share of firms reporting any product switching activity across the 1987 to 1997 censuses. Columns 3 and 4 report mean product add and drop rates. Columns 5 and 7 report the mean share of products' 1992 output due to firms that recently entered or are about to exit, respectively. Columns 6 and 8 report the mean share of products' 1992 output accounted for by firms that recently added or are about to drop the product, respectively. Columns 9 through 12 report decompositions similar to Columns 6 to 8 but with respect to the average share of firms producing a product in year t.

Table 1: Five-Digit SIC Products in Four-Digit SIC Industry 3357

SIC	Description	
33	Primary Metal Industries	
3357	Nonferrous Wiredrawing and Insulating	
33571	Products	Aluminum Wire
33572		Copper Wire
33573		Other Nonferrous Metal Wire
33575		Nonferrous Wire Cloth
33576		Apparatus Wire and Cord Sets
33577		Magnet Wire
33578		Power Wire
3357A		Electronic Wire
3357B		Telephone Wire
3357C		Control Wire
3357D		Building Wire
3357E		Other Wire NES
33579		Fiber Optic Cable

Source: U.S. Census Bureau (1989).

Table 2: 1997 Products per Industry and Product Characteristics, by Sector

			Products per	Capital Intensity		Skill Intensity	
Sector	Industries	Products	Industry	Mean	Std Dev	Mean	Std Dev
20 Food	49	156	3.2	169	127	0.21	0.11
22 Textile	23	82	3.4	171	220	0.18	0.11
23 Apparel	31	76	2.4	74	70	0.19	0.13
24 Lumber	17	59	3.5	273	158	0.19	0.06
25 Furniture	13	37	2.8	52	39	0.20	0.09
26 Paper	17	55	3.2	293	145	0.21	0.06
27 Printing & Publishing	14	70	5.1	102	97	0.48	0.27
28 Chemicals	29	100	3.5	438	359	0.34	0.13
29 Petroleum	5	16	3.0	878	418	0.34	0.08
30 Rubber & Plastics	15	61	4.2	185	205	0.26	0.10
31 Leather	11	12	1.1	52	23	0.20	0.10
32 Stone & Concrete	26	47	1.8	206	169	0.21	0.10
33 Primary Metal	26	90	3.4	176	109	0.22	0.08
34 Fabricated Metal	38	131	3.6	125	95	0.26	0.13
35 Industrial Machinery	51	178	3.7	123	80	0.29	0.16
36 Electronic	37	111	3.0	140	153	0.30	0.16
37 Transportation	18	61	3.6	128	70	0.31	0.18
38 Instruments	17	45	2.6	142	96	0.35	0.17
39 Miscellaneous	18	53	2.9	103	76	0.28	0.11
Total or average	455	1440	3.2				

Notes: Table reports the number of four-digit SIC industries and five-digit SIC products within the noted two-digit SIC major group, respectively, according to the 1987 revision of the SIC (Census 1989). Data on product capital and skill intensity are for 1997. Capital and skill intensity are the book value ($000) of buildings and equipment per worker, and non-production workers per total employment, respectively. Mean and standard deviation input intensities are computed across all plants producing a product within the relevant sector. Means are weighted by plant-product output. Results for Tobacco (SIC 21) are excluded due to Census disclosure guidelines.

Table 3: Prevalence of Firms Producing Multiple Products, Industries and Sectors in 1997

Type of Firm	Percent of Firms	Percent of Output	Mean Products, Industries or Sectors per Firm
Multiple Product	39	87	3.5
Multiple-Industry	28	81	2.8
Multiple-Sector	10	66	2.3

Notes: Table displays a breakdown of firms according to whether they produce multiple products (five-digit SIC categories), multiple industries (four-digit SIC categories) and multiple sectors (two-digit SIC categories). Columns 1 and 2 summarize the distribution of firms and output, respectively. Column 3 reports the mean number of products for multiple-product firms, the mean number of industries for multiple-industry firms and the mean number of sectors for multiple-sector firms. Results are based on the pooled 1987 to 1997 sample.

Table 4: 1997 MP versus SP Firm Characteristics

Firm Characteristic	Multiple Product	Multiple Industry	Multiple Sector
Output	0.66	0.67	0.92
Employment	0.58	0.61	0.86
Probability of Export	0.12	0.12	0.16
Labor Productivity	0.08	0.06	0.06
TFP	0.02	0.02	0.00

Notes: Table summarizes mean log differences in 1997 between firms that produce multiple products, industries and sectors versus those that produce a single product, industry and sector, respectively. Results are obtained by OLS regressions of the log of the noted firm characteristic on a dummy variable indicating the firms' status as well as industry fixed effects. A firm's industry is defined as the four-digit SIC in which it has the highest value of shipments. Regression standard errors are adjusted for clustering at the industry level. All regressions are restricted to the 110,414 observations for which all firm characteristics are available. All differences are statistically significant at the 1 percent level except for TFP premia for multi-sector firms.

Table 5: Product Switching by U.S. Manufacturing Firms, 1987 to 1997

	Percent of Firms				
Firm Activity	All Firms	Multi-Product Firms	Exporters	Large Firms	Multi-Plant Firms
None	46	20	38	39	25
Drop Product(s) Only	15	12	18	17	21
Add Product(s) Only	14	32	14	16	15
Both Add and Drop Products	25	36	31	28	38

	Output-Weighted Percent of Firms				
Firm Activity	All Firms	Multi-Product Firms	Exporters	Large Firms	Multi-Plant Firms
None	11	6	6	10	5
Drop Product(s) Only	10	8	9	10	10
Add Product(s) Only	10	12	9	10	9
Both Add and Drop Products	68	75	76	70	77

Note: Top panel displays average percent of surviving U.S. manufacturing firms engaging in each type of product-changing activity across five-year intervals from 1987 to 1997. Bottom panel provides a similar breakdown but weighting each firm by its output. Products refer to five-digit SIC categories. The four firm activities are mutually exclusive. "Large Firms" are defined as firms whose output is above the seventy-fifth percentile.

Table 6: Product Switching and Concomitant Changes in Firm Characteristics, 1987 to 1997

	Net Drop	Net Add	Obs	R^2
Real Output	-0.078 ***	0.096 ***	94,012	0.05
	0.009	0.008		
Employment	-0.085 ***	0.078 ***	94,012	0.03
	0.010	0.008		
Real Output/Worker	0.007 **	0.018 ***	94,012	0.03
	0.004	0.004		
TFP	-0.041 ***	0.031 ***	94,012	0.08
	0.007	0.008		

Notes: Table summarizes OLS regression results of log change in firm characteristics from 1987 to 1997 on dummy variables indicating whether firms net add or net drop products. The left out category is firms that undertake no product switching or, if they do switch products, experience no net change in the number of products they produce. Each row summarizes the regression for a different dependent variable. Columns report the two regression coefficients and their standard errors, the number of observations and the R-squared of the regression. Standard errors are adjusted for clustering at the product-mix level. Regressions include product mix by year fixed effects; coefficients for fixed effects and the constant are suppressed. Each regression is restricted to the set of observations for which all data are available. ***, **, and * represent statistical significance at the 1%, 5% and 10% levels, respectively.

Table 7: Sector and Industry Switching by U.S. Manufacturing Firms, 1987 to 1997

	Percent of Firms		
Firm Activity	Product Activity	Industry Activity	Sector Actvivity
None	46	59	84
Drop Only	15	14	6
Add Only	14	13	6
Both Add and Drop	25	14	3

Note: Table displays average share of surviving firms who engage in product, industry and sector switching across five-year intervals from 1987 to 1997. Product, industry and sector activity refers to the adding and/or dropping of five-digit, four-digit and two-digit SIC categories, respectively. The four firm activities are mutually exclusive.

Table 8: Average Decomposition of Product Output by Type of Producer, 1987 to 1997

Average Share (%) of Product Output in Year t Produced by:

	Backward-Looking			Forward-Looking		
	Firms Producing Product in Years t-5 and t	Firms that Add the Product Between Years t-5 and t	Firms Born Between Years t-5 and t	Firms Producing Product in Years t and t+5	Firms that Drop the Product Between Years t and t+5	Firms that Die Between Years t and t+5
1987	.	.	.	65	16	19
1992	67	14	19	67	15	18
1997	70	15	15	.	.	.

Average Share (%) of Firms Producing a Product in Year t by:

	Backward-Looking			Forward-Looking		
	Firms Producing Product in Years t-5 and t	Firms that Add the Product Between Years t-5 and t	Firms Born Between Years t-5 and t	Firms Producing Product in Years t and t+5	Firms that Drop the Product Between Years t and t+5	Firms that Die Between Years t and t+5
1987	.	.	.	44	27	29
1992	40	23	37	44	25	32
1997	45	26	29	.	.	.

Notes: Table reports average year t decomposition of product output (upper panel) and number of firms producing a product (lower panel) according to firm activity. Left-panel summarizes backward-looking firm activities while right panel summarizes forward-looking firm activities. Each row represents the average across all five-digit SIC products in the noted year. Decompositions cover the 1987 to 1997 censuses.

Table 9: 1992 to 1997 Product Adding OLS Regressions

	Single-Product Firms				Multiple-Product Firms			
	$Add_{t,t+5}$	$Add_{t,t+5}$	$Add_{t,t+5}$	$Add_{t,t+5}$	$Add_{t,t+5}$	$Add_{t,t+5}$	$Add_{t,t+5}$	$Add_{t,t+5}$
TFP_t	0.0140 ***	0.0150 ***			0.0020 ***	0.0026 ***		
	0.0031	0.0031			0.0049	0.0049		
$ln(Output/Worker)_t$			0.0099 ***	0.0114 ***			0.0178 ***	0.0137 ***
			0.0023	0.0023			0.0044	0.0044
$ln(Employment)_t$		0.0269 ***		0.0271 ***		0.0268 ***		0.0265 ***
		0.0012		0.0013		0.0022		0.0022
$ln(Age)_t$		0.0057 ***		0.0053 ***		0.0228 ***		0.0224 ***
		0.0021		0.0021		0.0036		0.0036
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Product-Mix Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R-Squared	0.07	0.07	0.07	0.07	0.51	0.52	0.51	0.52
Observations	105,035	105,036	105,037	105,038	74,976	74,977	74,978	74,979

Notes: Table summarizes OLS regression results of noted dummy variable indicating product adding by single-product (left panel) and multi-product (right panel) firms between 1992 and 1997. Robust standard errors are noted below each coefficient; standard errors are adjusted for clustering at the year t product-mix level. Coefficients for fixed effects and constant are suppressed. ***, **, and * represent statistical significance at the 1%, 5% and 10% levels, respectively.

Table 10: 1992 to 1997 Firm-Product OLS Drop Regressions

	Drop Product$_{t,t+5}$	Drop Product$_{t,t+5}$	Drop Product$_{t,t+5}$
Relative Product Size$_t$	-0.059 ***	-0.086 ***	-0.077 ***
	0.001	0.001	0.001
Relative Product Tenure	-0.189 ***	-0.219 ***	-0.223 ***
	0.006	0.008	0.008
Constant	0.256 ***	-0.001 ***	-0.071 ***
	0.003	0.001	0.002
Fixed Effects	None	Firm	Firm, Product
R-Squared	0.10	0.15	0.13
Observations	80,371	80,371	80,371

Notes: Table summarizes OLS regression results of dummy variable indicating a firm-product drop between 1992 and 1997 on 1992 firm-product attributes and firm and/or firm and product fixed effects. Firm-product size and tenure are relative to firms with the same mix of products. Regression restricted to surviving multiple-product firms. Robust standard errors are noted below each coefficient; standard errors are adjusted for clustering at the product level. Coefficients for fixed effects are suppressed. R-squared for final two columns is the "within" R-squared. ***, **, and * represent statistical significance at the 1%, 5% and 10% levels, respectively.

Table 11: Average Decomposition of Firm Output by Type of Product, 1987 to 1997

Average Share (%) of Firm Output in Year t Accounted for by:				
	Backward-Looking		Forward-Looking	
	Products Produced in Years t-5 and t	Products Added Between Years t-5 and t	Products Produced in Years t and t+5	Products DroppedBetween Years t and t+5
1987	.	.	71	29
1992	74	26	74	26
1997	69	31	.	.

Notes: Table reports average year t decomposition of firm output according to whether products were previously (right panel) or subsequently (left panel) produced. Each row represents the average across all adding or dropping firms in the noted year. Decompositions cover the 1987 to 1997 censuses.

Table 12: Mean Distribution of Within-Firm Output Shares, 1987 to 1997

Average Percent of Output	Number of Products Produced by the Firm									
	1	2	3	4	5	6	7	8	9	10
1	100	80	70	63	58	54	52	50	48	46
2		19	21	22	21	21	21	20	20	20
3			7	9	10	11	11	11	11	12
4				4	5	6	7	7	7	7
5					2	3	4	4	5	5
6						2	2	3	3	3
7							1	2	2	2
8								1	1	2
9									1	1
10										1

Notes: Columns indicate the number of products produced by the firm. Rows indicate the share of the produce, in descending order of size. Each cell is the average across the relevant set of firm-products in the sample. Sample includes all firms producing at least ten products in the 1987 to 1997 Censuses.

Table 13: Product Co-Production Within Firms, 1987 to 1997

Sector	20	31	30	22	23	39	24	26	25	27	28	29	32	33	34	35	36	37	38
20 Food	158	2	9	5	16	5	6	9	6	11	42	3	4	6	15	16	8	5	4
31 Leather	2	1	1	1	3	1	1	1	1	1	1	0	0	1	2	1	1	0	0
30 Rubber & Plastic	9	1	47	8	9	7	8	10	6	6	24	2	6	13	33	41	22	12	11
22 Textile	5	1	8	25	18	3	3	4	3	4	11	1	2	4	8	8	5	3	3
23 Apparel	16	3	9	18	64	6	6	6	6	9	12	1	3	7	20	21	9	5	5
39 Miscellaneous	5	1	7	3	6	10	3	3	2	9	6	0	1	3	10	9	6	3	3
24 Lumber	6	1	8	3	6	3	79	16	16	5	9	1	4	5	16	10	6	3	3
26 Paper	9	1	10	4	6	3	16	19	3	23	13	1	3	5	12	12	6	2	4
25 Furniture	6	1	6	3	6	2	16	3	23	3	4	0	1	4	12	10	7	3	3
27 Printing & Publishing	11	1	6	4	9	9	5	23	3	401	10	1	2	5	13	15	10	4	5
28 Chemicals	42	1	24	11	12	6	9	13	4	10	74	13	10	16	25	29	20	10	18
29 Petroleum	3	0	2	1	1	0	1	1	0	1	13	4	2	3	4	3	2	1	1
32 Stone & Concrete	4	0	6	2	3	1	4	3	1	2	10	2	10	6	11	11	7	3	3
33 Primary Metal	6	1	13	4	7	3	5	5	4	5	16	3	6	27	48	47	29	15	10
34 Fabricated Metal	15	2	33	8	20	10	16	12	12	13	25	4	11	48	124	117	53	30	22
35 Industrial Machinery	16	1	41	8	21	9	10	12	10	15	29	3	11	47	117	119	72	38	31
36 Electronic	8	1	22	5	9	6	6	6	7	10	20	2	7	29	53	72	70	25	37
37 Transportation	5	0	12	3	5	3	3	2	3	4	10	1	3	15	30	38	25	18	10
38 Instruments	4	0	11	3	5	3	3	4	3	5	18	1	3	10	22	31	37	10	14

Notes: Table summarizes co-production of five-digit SIC categories across multiple-product firms. Each cell contains a count (in 000s) of the average number of firms observed producing two products in the noted two-digit SIC sector across the 1987 to 1997 Census years. Firms with more than two products may be counted more than once in each cell. Dark and light grey shading indicate statistically significantly higher and lower co-production at the 1 percent, respectively, than is implied by a null hypothesis of random co-production (see text). The absence of shading indicates the absence of statistically significant differences from this null hypothesis. Sectors have been sorted to maximize dark shading along the diagonal (see text).

Table 14: How Products are Added and Dropped, 1987 to 1997

	Share of Products		Share of Firms	
Method of Product Adding	Unweighted	Value-Weighted	Unweighted	Value-Weighted
Existing Plant(s) Only	0.862	0.412	0.899	0.259
Acquired Plant(s) Only	0.055	0.259	0.013	0.048
New Plant(s) Only	0.060	0.120	0.030	0.044
Combination with M&A	0.013	0.147	0.031	0.488
Combination without M&A	0.011	0.061	0.027	0.161

	Share of Products		Share of Firms	
Method of Product Dropping	Unweighted	Value-Weighted	Unweighted	Value-Weighted
Existing Plant(s) Only	0.835	0.320	0.898	0.213
Divested Plant(s) Only	0.052	0.282	0.007	0.018
Closed Plants Only	0.084	0.123	0.037	0.042
Combination with M&A	0.015	0.231	0.026	0.538
Combination without M&A	0.013	0.045	0.033	0.189

Notes: Table reports the manner in which firms add (top panel) and drop (bottom panel) five-digit SIC products. The first two columns report the distribution with respect to products; the second two columns report the distribution with respect to firms. Figures shown are averages across the pooled 1987 to 1997 sample.

Table 15: Product Switching and Concurrent Merger and Acquisition Activity, 1987 to 1997

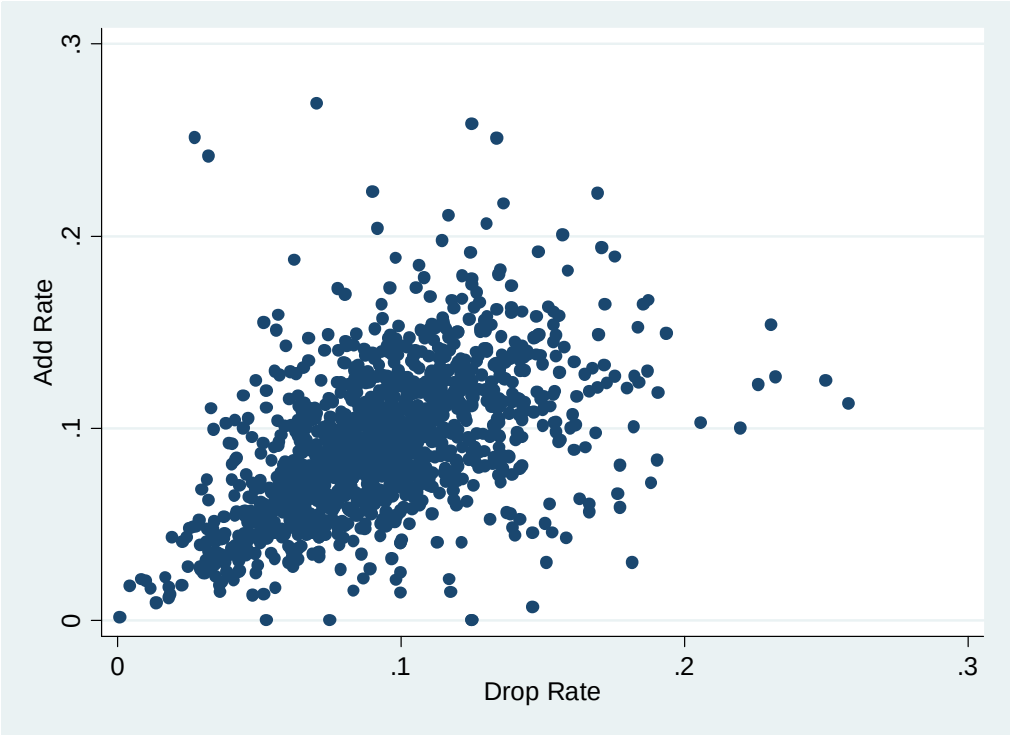
	Product Switching		Sector Switching	
Firm Activity	No M&A	With M&A	No M&A	With M&A
None	47	6	85	33
Drop Only	15	20	6	25
Add Only	14	14	6	26
Both Add and Drop	24	60	3	17

Notes: Table displays average percent of surviving U.S. manufacturing firms engaging in each type of product- (left panel) or sector- (right panel) changing activity across five-year intervals from 1987 to 1997 according to whether or not they concomitantly engage in any M&A activity (i.e., acquire or divest plants). Products refer to five-digit SIC categories while sectors refer to two-digit SIC categories. The four firm activities are mutually exclusive.

Table 16: Two-digit SIC Sector Results

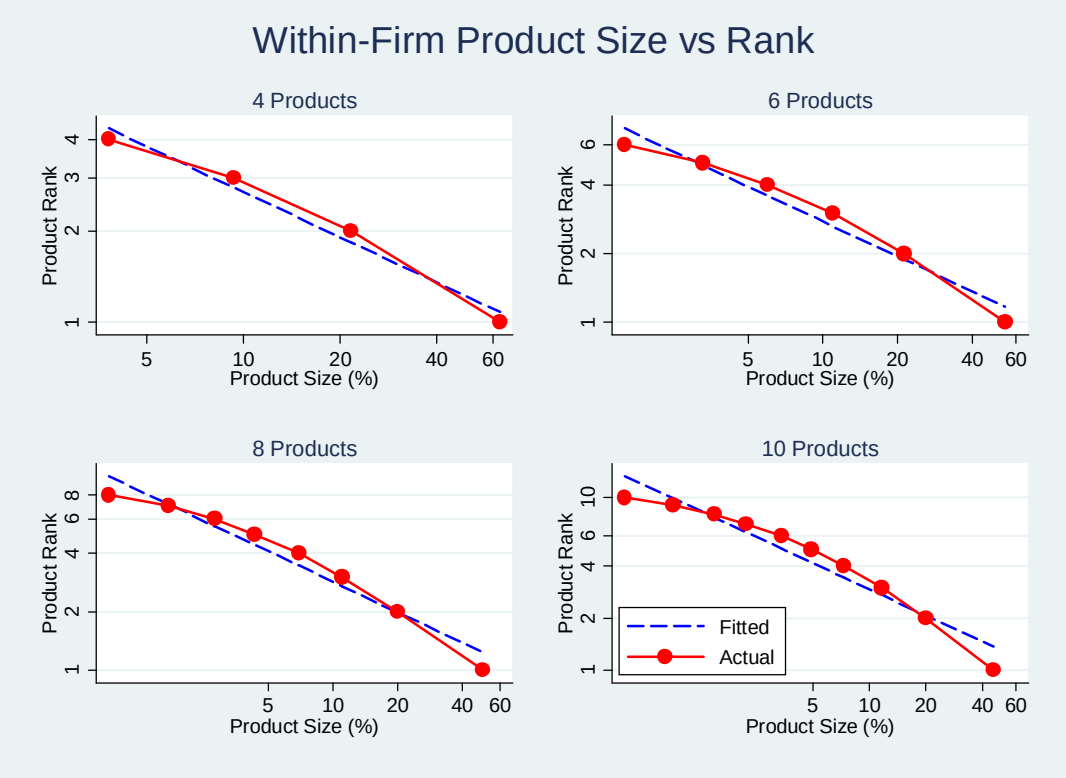
Sector	MP Firms (Table 3)	Product Switching (Table 5)	Add Rate (Figure 1)	Drop Rate (Figure 1)	Entering Firms' Output Share (1992; Table 8)	Adding Firms' Output Share (1992; Table 8)	Exiting Firms' Output Share (1992; Table 8)	Dropping Firms' Output Share (1992; Table 8)	Entering Firms (%) (1992; Table 8)	Adding Firms (%) (1992; Table 8)	Exiting Firms (%) (1992; Table 8)	Dropping Firms (%) (1992; Table 8)
20 Food	48	51	8	8	12	10	13	12	30	20	28	20
22 Textile	33	53	9	9	16	23	21	20	32	26	35	24
23 Apparel	22	47	8	7	25	16	29	18	44	20	48	18
24 Lumber	38	53	9	10	28	17	20	16	43	22	35	22
25 Furniture	42	60	9	9	22	12	20	18	48	21	44	22
26 Paper	32	42	8	9	14	10	16	13	27	19	24	22
27 Printing & Publishing	60	71	11	10	21	16	25	15	45	26	36	30
28 Chemicals	46	59	10	10	13	12	11	14	30	23	26	27
29 Petroleum	28	34	6	7	14	7	9	7	27	14	24	16
30 Rubber & Plastics	42	65	11	10	18	23	16	16	36	28	28	26
31 Leather	26	39	5	6	22	7	31	12	44	11	45	14
32 Stone & Concrete	22	33	7	8	18	12	14	15	38	17	31	18
33 Primary Metal	46	53	9	9	19	14	17	13	30	25	26	25
34 Fabricated Metal	37	52	11	11	22	17	19	17	38	26	29	30
35 Industrial Machinery	30	49	10	10	23	14	18	18	39	24	30	29
36 Electronic	33	52	10	10	16	16	14	14	37	25	30	28
37 Transportation	40	59	9	10	18	11	17	20	37	24	31	28
38 Instruments	28	52	8	9	16	15	16	16	43	22	35	26
39 Miscellaneous	34	46	7	7	24	13	24	11	43	16	39	18
All	38	54	8	9	19	14	18	15	37	23	32	25

Note: This table summarizes the paper's main findings across two-digit SIC industries. Table in which main results can be found is noted in parentheses at the top of each column. Column 1 reports the mean share of multi-product firms across the 1987 to 1997 censuses. Column 2 reports the mean share of firms reporting any product switching activity across the 1987 to 1997 censuses. Columns 3 and 4 report mean product add and drop rates. Columns 5 and 7 report the mean share of products' 1992 output due to firms that recently entered or are about to exit, respectively. Columns 6 and 8 repot the mean share of products' 1992 output accounted for by firms that recently added or are about to drop the product, respectively. Columns 9 through 12 report decompositions similar to columns 6 to 8 but with respect to the average share of firms producing a product in year t. Columns 5 through 12 report data for 1992.



Notes: Figure displays average rates at which five-digit SIC products are added and dropped from 1987 to 1997. A product's add (drop) rate is defined as the number of firms adding (dropping) the product between census years divided by the average number of firms producing the product in both years.

Figure 1: Average Product Add and Drop Rates, 1987 to 1997



Notes: Vertical and horiztonal axes use log scales. The solid line in each panel plots within-firm product rank against within-firm product size. Each panel is for the set of firms with the noted number of products. The largest product has a rank of one. A product's size is defined as its mean percent of firm shipments, as noted in Table 11. Fitted lines are the result of an ordinary-least-squares regression of log product rank on log product size.

Figure 2: Distribution of Product Shipments Within Firms, 1987 to 1997